

Circuit-Based Quantum Computing

(Session 2 – Day 1, 3 July 2026)

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Learning Objectives

- Understand the quantum circuit model and its correspondence to unitary operations on Hilbert spaces.
- Become familiar with the elementary single-qubit and multi-qubit gates, including their matrix representations.
- Learn to read, construct, and draw quantum circuit diagrams (using the `quantikz` package).
- Work through key algorithms and protocols step-by-step: Bell state preparation, superdense coding, and the Deutsch algorithm.
- Grasp the concept of universality, gate sets (Clifford+ T , Toffoli+ H), and the Solovay–Kitaev theorem.

1 From States to Circuits

A quantum algorithm in the circuit model is a sequence of discrete unitary operations (gates) applied to a register of qubits initialized in a standard state (usually $|0\rangle^{\otimes n}$), followed by measurement in the computational basis.

Mathematically, if $U_1, U_2, \dots, U_m$ represent the sequential gates applied to the system, the overall computation is described by a single unitary matrix:

$$U = U_m \dots U_2 U_1$$

If the initial state of the register is $|\psi_0\rangle = |0\rangle^{\otimes n}$, the final state before measurement is:

$$|\psi_f\rangle = U |0\rangle^{\otimes n}$$

Measurements are then performed in the computational basis $\{|x\rangle \mid x \in \{0, 1\}^n\}$ to extract classical information.

2 Elementary Gates

Below are the mathematical matrix representations of the standard gates in quantum computing:

Single-qubit gates (unitary operations on $\mathbb{C}^2$):

$$\begin{aligned} X &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, & Y &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, & Z &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ H &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, & S &= \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, & T &= \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \end{aligned}$$

Note that X , Y , and Z are the Pauli matrices, representing π rotations about the x , y , and z axes of the Bloch sphere, respectively. H is the Hadamard gate, and $T = \sqrt{S} = \sqrt[4]{Z}$ is the $\pi/8$ phase gate.

Multi-qubit gates (unitary operations on $\mathbb{C}^4$ and $\mathbb{C}^8$):

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X$$

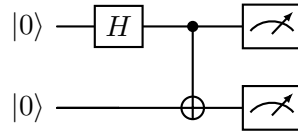
$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = |00\rangle\langle 00| + |01\rangle\langle 10| + |10\rangle\langle 01| + |11\rangle\langle 11|$$

$$\text{Toffoli (CCNOT)} = |00\rangle\langle 00| \otimes I + |01\rangle\langle 01| \otimes I + |10\rangle\langle 10| \otimes I + |11\rangle\langle 11| \otimes X$$

3 Circuit Diagrams with quantikz

A quantum circuit diagram is read from left to right. Each horizontal wire represents a single qubit. Single-qubit gates are represented by labeled boxes, and multi-qubit gates are shown by connecting vertical lines.

The circuit below preparing the Bell state $|\Phi^+\rangle$ is drawn as:



4 Toy Example 1: Bell State Preparation

The Bell states (or EPR pairs) are maximally entangled 2-qubit states. We prepare the canonical Bell state:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

The preparation steps are:

1. Start with the product state $|\psi_0\rangle = |00\rangle$.
2. Apply a Hadamard gate to qubit 0:

$$|\psi_1\rangle = (H \otimes I) |00\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes |0\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

3. Apply a Controlled-NOT (CNOT) gate with qubit 0 as the control and qubit 1 as the target:

$$|\psi_2\rangle = \text{CNOT} \left(\frac{|00\rangle + |10\rangle}{\sqrt{2}} \right) = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\Phi^+\rangle$$

If both qubits are measured in the computational basis, the outcomes are perfectly correlated: either 00 or 11, each with a probability of 50%.

5 Toy Example 2: Superdense Coding

Superdense coding is a protocol showing how a sender (Alice) can transmit two classical bits ($ab \in \{00, 01, 10, 11\}$) to a receiver (Bob) by physically sending only a single qubit, given a pre-shared Bell pair.

5.1 Protocol Steps

1. **Pre-shared Resource:** Alice and Bob share the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$. Alice holds qubit A, and Bob holds qubit B.
2. **Alice's Encoding:** Depending on the 2 classical bits ab she wants to send, Alice applies local unitary operations to her qubit A:

- **If $ab = 00$:** Alice applies I to qubit A. State remains:

$$|\psi_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\Phi^+\rangle$$

- **If $ab = 01$:** Alice applies X to qubit A. State becomes:

$$|\psi_{01}\rangle = (X \otimes I) |\Phi^+\rangle = \frac{|10\rangle + |01\rangle}{\sqrt{2}} = |\Psi^+\rangle$$

- **If $ab = 10$:** Alice applies Z to qubit A. State becomes:

$$|\psi_{10}\rangle = (Z \otimes I) |\Phi^+\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} = |\Phi^-\rangle$$

- **If $ab = 11$:** Alice applies XZ (or iY) to qubit A. State becomes:

$$|\psi_{11}\rangle = (XZ \otimes I) |\Phi^+\rangle = \frac{|10\rangle - |01\rangle}{\sqrt{2}} = -|\Psi^-\rangle$$

3. **Transmission:** Alice physically sends her single qubit A to Bob.
4. **Bob's Decoding:** Bob now holds both qubits A and B. He applies a joint Bell-basis measurement (a CNOT gate from A to B, followed by a Hadamard gate on A, then measures both):

- Bob applies CNOT from A to B. For instance, on $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$:

$$\text{CNOT}|\Psi^+\rangle = \frac{|01\rangle + |11\rangle}{\sqrt{2}} = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)|1\rangle = |+\rangle|1\rangle$$

- Bob applies H to qubit A:

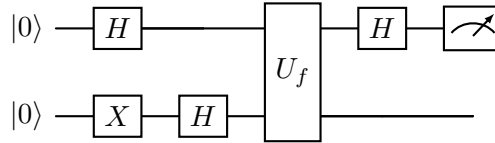
$$(H \otimes I)|+\rangle|1\rangle = |0\rangle|1\rangle = |01\rangle$$

- Measuring both qubits yields the exact bit string 01 with 100% certainty.

6 Toy Example 3: Deutsch Algorithm

The Deutsch algorithm is the simplest algorithm demonstrating a quantum speedup. Given a black-box function $f : \{0, 1\} \rightarrow \{0, 1\}$ (an oracle), we are promised that f is either **constant** ($f(0) = f(1)$) or **balanced** ($f(0) \neq f(1)$). We want to determine its classification using a single query to the oracle, whereas a classical algorithm requires evaluating both $f(0)$ and $f(1)$ (two queries).

The quantum circuit using **phase kickback** is shown below:



Here, the black box U_f is defined as the unitary transformation acting on two qubits as:

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

6.1 Step-by-Step Mathematical Analysis

1. **Initial State:** Qubits are initialized to $|\psi_0\rangle = |0\rangle|0\rangle$.
2. **Prepare Superposition:** We apply X and H to qubit 1 (target) to prepare it in the state $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$, and apply H to qubit 0 (control) to prepare it in the state $|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$:

$$|\psi_1\rangle = (H \otimes HX)|\psi_0\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

3. **Derivation of Phase Kickback:** We apply the oracle U_f , which acts as $U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$. Let us evaluate its action when the target qubit is in the superposition state $|-\rangle$:

$$\begin{aligned} U_f |x\rangle |-\rangle &= U_f |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= \frac{U_f |x\rangle |0\rangle - U_f |x\rangle |1\rangle}{\sqrt{2}} \\ &= \frac{|x\rangle |0 \oplus f(x)\rangle - |x\rangle |1 \oplus f(x)\rangle}{\sqrt{2}} \end{aligned}$$

Depending on the value of $f(x) \in \{0, 1\}$, we have two cases:

- **If $f(x) = 0$:**

$$\frac{|x\rangle |0\rangle - |x\rangle |1\rangle}{\sqrt{2}} = |x\rangle |-\rangle = (-1)^0 |x\rangle |-\rangle$$

- **If $f(x) = 1$:**

$$\frac{|x\rangle |1\rangle - |x\rangle |0\rangle}{\sqrt{2}} = - \left(\frac{|x\rangle |0\rangle - |x\rangle |1\rangle}{\sqrt{2}} \right) = - |x\rangle |-\rangle = (-1)^1 |x\rangle |-\rangle$$

Thus, for any $f(x)$, we have:

$$U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle$$

The query value $f(x)$ is “kicked back” as a phase factor $(-1)^{f(x)}$ onto the control qubit.

4. **Applying the Oracle to the Joint State:** Applying U_f to the state $|\psi_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle |-\rangle + |1\rangle |-\rangle)$:

$$|\psi_2\rangle = U_f |\psi_1\rangle = \frac{1}{\sqrt{2}} (U_f |0\rangle |-\rangle + U_f |1\rangle |-\rangle) = \left(\frac{(-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle}{\sqrt{2}} \right) \otimes |-\rangle$$

5. **Analysis for a Balanced Function ($f(0) \neq f(1)$):** Since f is balanced, the two values $f(0)$ and $f(1)$ must be different. There are two sub-cases:

- **Case A ($f(0) = 0, f(1) = 1$):**

$$|\psi_2\rangle = \left(\frac{(-1)^0 |0\rangle + (-1)^1 |1\rangle}{\sqrt{2}} \right) \otimes |-\rangle = \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \otimes |-\rangle = |-\rangle |-\rangle$$

- **Case B ($f(0) = 1, f(1) = 0$):**

$$|\psi_2\rangle = \left(\frac{(-1)^1 |0\rangle + (-1)^0 |1\rangle}{\sqrt{2}} \right) \otimes |-\rangle = \left(\frac{-|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes |-\rangle = -|-\rangle |-\rangle$$

Therefore, if f is balanced, the joint state before the final Hadamard gate is $\pm |-\rangle |-\rangle$.

6. **Final Rotation and Measurement:** We apply $H \otimes I$ to $|\psi_2\rangle = \pm |-\rangle |-\rangle$. Since $H |-\rangle = |1\rangle$, the final state is:

$$|\psi_3\rangle = \pm |1\rangle |-\rangle$$

Measuring the first (control) qubit in the computational basis yields **1 with 100% probability**.

7 Universality

A set of gates is said to be **universal** if any unitary operation on any number of qubits can be approximated to arbitrary precision by a finite sequence of gates from that set.

Common universal gate sets include:

- **Clifford + T Set:** $\{H, S, T, \text{CNOT}\}$. This is the standard set for fault-tolerant quantum computing. Note that $S = T^2$, so $\{H, T, \text{CNOT}\}$ is also universal.
- **Toffoli + Hadamard Set:** $\{\text{Toffoli}, H\}$. The Toffoli gate is universal for classical reversible circuits, and adding the Hadamard gate enables quantum superposition, making it universal for quantum computing.

The **Solovay–Kitaev theorem** guarantees that any single-qubit unitary can be compiled into a sequence of Clifford+ T gates of length $\mathcal{O}(\log^c(1/\epsilon))$ to achieve an approximation error of ϵ , where $c \approx 3.97$.

8 Clifford and Non-Clifford Operations

In fault-tolerant quantum computing (FTQC), quantum gates are categorized into two classes based on their ease of implementation and classical simulability: **Clifford operations** and **Non-Clifford operations**.

8.1 The Clifford Group

The n -qubit Clifford group $\mathcal{C}_n$ is defined as the normalizer of the n -qubit Pauli group $\mathcal{P}_n$ within the unitary group $\mathcal{U}(2^n)$:

$$\mathcal{C}_n = \{U \in \mathcal{U}(2^n) \mid UPU^\dagger \in \mathcal{P}_n \text{ for all } P \in \mathcal{P}_n\}$$

In other words, Clifford unitaries map Pauli operators to other Pauli operators under conjugation. The Clifford group is generated by three elementary gates:

- The **Hadamard gate** (H): Maps $X \rightarrow Z$ and $Z \rightarrow X$.
- The **Phase gate** (S): Maps $X \rightarrow Y$ and $Z \rightarrow Z$.
- The **Controlled-NOT gate** (CNOT): Maps $X \otimes I \rightarrow X \otimes X$ and $I \otimes Z \rightarrow Z \otimes Z$.

Stabilizer states are defined as states that can be prepared from $|0\rangle^{\otimes n}$ using only Clifford gates.

The Gottesman–Knill Theorem

Any quantum circuit consisting solely of:

1. Stabilizer state preparations (inputs in the computational basis),
2. Clifford group gates (H , S , CNOT), and
3. Measurements in the Pauli bases

can be efficiently simulated on a classical deterministic computer in polynomial time ($\mathcal{O}(n^2)$ classical operations per gate/measurement).

Therefore, Clifford operations alone cannot provide any quantum speedup or computational advantage.

8.2 Non-Clifford Operations and Magic States

To achieve universality and quantum computational advantage, we must introduce at least one non-Clifford gate. The standard choice is the T **gate** (or $\pi/8$ phase gate):

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

Because T is not in the Clifford group, adding T to the Clifford group allows us to approximate any unitary to arbitrary accuracy (universality).

However, in fault-tolerant architectures (such as surface codes), Clifford gates can be implemented transversally (meaning they are naturally protected against errors), whereas non-Clifford gates like T cannot be implemented transversally due to the Eastin–Knill theorem. Instead, they must be performed using a technique called **magic state distillation and injection**.

8.3 Magic State Injection

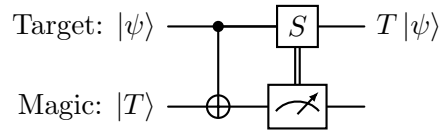
Instead of applying the non-Clifford T gate directly to our target state $|\psi\rangle$, we can prepare an auxiliary resource state, known as the **magic state** $|T\rangle$, offline:

$$|T\rangle = T|+\rangle = \frac{|0\rangle + e^{i\pi/4}|1\rangle}{\sqrt{2}}$$

We then “inject” this magic state into our target qubit using only Clifford gates, measurements, and classical feedforward control (correction).

8.3.1 Circuit Diagram

The circuit for T -gate injection is shown below:



Here, the double line (classical wire) carries the measurement outcome $m \in \{0, 1\}$ to conditionally apply the Clifford Phase gate S as a correction.

8.3.2 Mathematical Verification

Let the target state be $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

1. The initial joint state of the system is:

$$|\Psi_0\rangle = |\psi\rangle \otimes |T\rangle = \frac{1}{\sqrt{2}}(\alpha|0\rangle + \beta|1\rangle) \otimes (|0\rangle + e^{i\pi/4}|1\rangle)$$

$$|\Psi_0\rangle = \frac{1}{\sqrt{2}} \left(\alpha|00\rangle + \alpha e^{i\pi/4}|01\rangle + \beta|10\rangle + \beta e^{i\pi/4}|11\rangle \right)$$

2. Apply a Controlled-NOT (CNOT) gate with the target qubit (top) as control and the magic state (bottom) as target:

$$|\Psi_1\rangle = \text{CNOT} |\Psi_0\rangle = \frac{1}{\sqrt{2}} \left(\alpha |00\rangle + \alpha e^{i\pi/4} |01\rangle + \beta |11\rangle + \beta e^{i\pi/4} |10\rangle \right)$$

Rearranging terms to group by the state of the bottom qubit:

$$|\Psi_1\rangle = \frac{1}{\sqrt{2}} \left[\left(\alpha |0\rangle + \beta e^{i\pi/4} |1\rangle \right) \otimes |0\rangle + \left(\alpha e^{i\pi/4} |0\rangle + \beta |1\rangle \right) \otimes |1\rangle \right]$$

3. Perform a Z -basis measurement on the bottom qubit:

- **Outcome** $m = 0$: The top qubit collapses to the state:

$$|\psi_{\text{out},0}\rangle = \alpha |0\rangle + \beta e^{i\pi/4} |1\rangle = T |\psi\rangle$$

This is exactly the desired state. No correction is needed.

- **Outcome** $m = 1$: The top qubit collapses to the state:

$$|\psi_{\text{out},1}\rangle = \alpha e^{i\pi/4} |0\rangle + \beta |1\rangle = e^{i\pi/4} \left(\alpha |0\rangle + \beta e^{-i\pi/4} |1\rangle \right) = e^{i\pi/4} T^\dagger |\psi\rangle$$

To correct this, we apply the Clifford phase gate $S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$:

$$S |\psi_{\text{out},1}\rangle = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} \alpha e^{i\pi/4} \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha e^{i\pi/4} \\ i\beta \end{pmatrix} = e^{i\pi/4} \left(\alpha |0\rangle + e^{i\pi/4} \beta |1\rangle \right) = e^{i\pi/4} T |\psi\rangle$$

This is the desired state $T |\psi\rangle$ up to an unobservable global phase of $e^{i\pi/4}$.

Thus, using one magic state $|T\rangle$, a Clifford CNOT gate, a measurement, and a conditional Clifford correction S , we can implement the non-Clifford T gate deterministically.

9 Circuit Decomposition Example (Z-Y-Z Euler Angles)

Any single-qubit unitary U can be decomposed as a product of rotations about the y and z axes:

$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta)$$

For instance, the Hadamard gate H can be decomposed up to a global phase of $e^{i\pi/2} = i$ as:

$$H = i R_z(\pi) R_y(\pi/2) R_z(\pi)$$

10 Discussion Problems / Exercises

For in-class discussion and problem-solving, consider the following exercises:

1. **CNOT gate decomposition:** Show how to construct a Controlled-NOT (CNOT) gate using only a Controlled-Z (CZ) gate and single-qubit Hadamard gates.

Session Takeaway: A CNOT gate can be rotated into a CZ gate using Hadamard gates on the target qubit: $\text{CNOT} = (I \otimes H)\text{CZ}(I \otimes H)$. This is because $HXH = Z$.

2. **Toffoli (CCNOT) gate matrix construction:** Write down the explicit action of the Toffoli gate on the 8 basis states of a 3-qubit system.

Session Takeaway: The Toffoli gate flips the third qubit if and only if both the first and second qubits are in state $|1\rangle$. It preserves all other basis states, making it a reversible classical AND gate.

3. **Deutsch oracle verification:** Show step-by-step how the oracle U_f acts on the state $|+\rangle|-\rangle$ for the balanced function $f(x) = x$ to produce the phase kickback.

Session Takeaway: For $f(x) = x$, the oracle acts as a CNOT gate. Applying CNOT to $|+\rangle|-\rangle$ flips the target state $|-\rangle$ conditioned on the control qubit, which mathematically shifts the relative phase of the control qubit, yielding the state $|-\rangle|-\rangle$.

11 Take-away Messages

1. Quantum circuits provide a modular, standardized representation of quantum algorithms using wires and unitary blocks.
2. Universal gate sets (like Clifford+ T) show that a small set of simple operations is sufficient to construct any complex quantum algorithm.
3. Fundamental protocols (Bell preparation, superdense coding, Deutsch algorithm) demonstrate how quantum mechanics allows us to manipulate information in ways impossible classically.

Further reading (optional)

Nielsen & Chuang, Chap. 4;

Qiskit Textbook, “Quantum Circuits and Gates”;

PennyLane documentation, “Operations”.

A Detailed Solutions to Discussion Problems

A.1 Solution to Problem 1: CNOT from CZ and Hadamard

We wish to verify the relation:

$$\text{CNOT} = (I \otimes H)\text{CZ}(I \otimes H)$$

Let's analyze the action of the right-hand side operator on the computational basis states:

1. For the state $|00\rangle$:

$$(I \otimes H) |00\rangle = |0\rangle |+\rangle$$

Applying CZ (which does nothing to the state if the control qubit is $|0\rangle$):

$$\text{CZ} |0\rangle |+\rangle = |0\rangle |+\rangle$$

Applying the second $(I \otimes H)$ gate:

$$(I \otimes H) |0\rangle |+\rangle = |00\rangle$$

2. For the state $|01\rangle$:

$$(I \otimes H) |01\rangle = |0\rangle |-\rangle \xrightarrow{\text{CZ}} |0\rangle |-\rangle \xrightarrow{I \otimes H} |01\rangle$$

3. For the state $|10\rangle$:

$$(I \otimes H) |10\rangle = |1\rangle |+\rangle = \frac{|10\rangle + |11\rangle}{\sqrt{2}}$$

Applying CZ applies a phase of -1 only to the state $|11\rangle$:

$$\text{CZ} \left(\frac{|10\rangle + |11\rangle}{\sqrt{2}} \right) = \frac{|10\rangle - |11\rangle}{\sqrt{2}} = |1\rangle |-\rangle$$

Applying the second $(I \otimes H)$ gate:

$$(I \otimes H) |1\rangle |-\rangle = |11\rangle$$

4. For the state $|11\rangle$:

$$(I \otimes H) |11\rangle = |1\rangle |-\rangle = \frac{|10\rangle - |11\rangle}{\sqrt{2}}$$

Applying CZ:

$$\text{CZ} \left(\frac{|10\rangle - |11\rangle}{\sqrt{2}} \right) = \frac{|10\rangle + |11\rangle}{\sqrt{2}} = |1\rangle |+\rangle$$

Applying the second $(I \otimes H)$ gate:

$$(I \otimes H) |1\rangle |+\rangle = |10\rangle$$

Comparing these actions with the CNOT gate:

$$\begin{array}{ll} |00\rangle \rightarrow |00\rangle, & |01\rangle \rightarrow |01\rangle \\ |10\rangle \rightarrow |11\rangle, & |11\rangle \rightarrow |10\rangle \end{array}$$

We see that the actions match exactly, proving the decomposition.

A.2 Solution to Problem 2: Toffoli Gate Basis Actions

The Toffoli (CCNOT) gate acts on three qubits. If the control qubits (the first two qubits) are both in state $|1\rangle$, the target qubit (the third qubit) is flipped. Otherwise, it does nothing. The actions on the 8 basis states are:

$$\begin{aligned} \text{CCNOT } |000\rangle &= |000\rangle, & \text{CCNOT } |001\rangle &= |001\rangle \\ \text{CCNOT } |010\rangle &= |010\rangle, & \text{CCNOT } |011\rangle &= |011\rangle \\ \text{CCNOT } |100\rangle &= |100\rangle, & \text{CCNOT } |101\rangle &= |101\rangle \\ \text{CCNOT } |110\rangle &= |111\rangle, & \text{CCNOT } |111\rangle &= |110\rangle \end{aligned}$$

A.3 Solution to Problem 3: Phase Kickback Validation for $f(x) = x$

For $f(x) = x$, the oracle is $U_f |x\rangle |y\rangle = |x\rangle |y \oplus x\rangle$, which is exactly a CNOT gate with control 0 and target 1. We analyze the action of U_f on the initial superposition state $|\psi_1\rangle = |+\rangle |-\rangle$:

$$|\psi_1\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

Applying CNOT (U_f):

$$U_f |\psi_1\rangle = \frac{1}{2} (\text{CNOT } |00\rangle - \text{CNOT } |01\rangle + \text{CNOT } |10\rangle - \text{CNOT } |11\rangle)$$

$$U_f |\psi_1\rangle = \frac{1}{2} (|00\rangle - |01\rangle + |11\rangle - |10\rangle)$$

Rearranging terms to factor the state:

$$U_f |\psi_1\rangle = \frac{1}{2} (|0\rangle (|0\rangle - |1\rangle) - |1\rangle (|0\rangle - |1\rangle)) = \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |-\rangle |-\rangle$$

Notice that:

- The target qubit remains in the state $|-\rangle$.
- The control qubit has evolved from state $|+\rangle$ to state $|-\rangle$.

This demonstrates that the action of the gate has kicked back a phase of -1 onto the control qubit when it was in state $|1\rangle$, changing its state from $|+\rangle$ to $|-\rangle$.

B List of Acronyms

Acronym	Description
CNOT	Controlled-NOT
CCNOT	Controlled-Controlled-NOT (also known as the Toffoli gate)
CZ	Controlled-Z (Controlled-Phase)
EPR	Einstein–Podolsky–Rosen (as in EPR pairs or Bell states)
FTQC	Fault-Tolerant Quantum Computing
SK	Solovay–Kitaev (relating to single-qubit gate approximation)